

The Phenomenology of the Anger Spectrum in the Qur'an

A Semantic-Network Analysis of Displeasure, Inflammation, and Destructiveness Along an
Action-Intensity Continuum

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April 2026

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Abstract

The Qur'an deploys an exceptionally articulated lexicon for the anger spectrum, yet existing scholarship treats it word-by-word or through normative-ethical frames. This study argues that fourteen core Arabic roots (ʾff, krh, dyq, hzn, ʾsf, nqm, sxt, mqt, ghdb, hrd, ghyz, bghy, tghy, ʾtw) form a graded semantic field organised by intensity of action across six phenomenological stages: pre-anger displeasure, inner pressure, evaluative aversion, active anger, compressed rage, and behavioural outcomes. The framework integrates Izutsu's semantic-field method, Kennedy–Sassoon scalar semantics, and Lakoff–Kövecses conceptual metaphor theory, distinguishing primary (ANGER IS HEAT) from complex (PRESSURIZED-CONTAINER ANGER) metaphors. We validate the continuum against the Quranic Arabic Corpus through a reproducible Python pipeline over 312 attestations. Three findings emerge under uncertainty-aware metrics. First, asymptotic $\chi^2(5) = 227.15$ is large, but a marginal-preserving permutation null tests which six bins absorb the baghy mass and yields $p = 0.24$. Second, three roots survive Holm–Bonferroni correction over fourteen per-root binomials: karh (Medinan, adj. $p \approx 10^{-6}$), baghy (Meccan, adj. $p \approx 5 \times 10^{-3}$), and ghayz (Medinan, adj. $p \approx 2 \times 10^{-2}$); sakhat sits at adj. $p = 0.05$. The pattern instantiates discursive-contextual lexical specialisation. Third, network analysis identifies baghy as a bridging node to the moral-evaluation field; Q. 67:8 of sūrat al-Mulk (takādu tamayyazu min al-ghayz)

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exemplifies the construal of vessel-internal failure newly available under CMT; a $\kappa = 0.79$ -validated tafsīr audit supports a bidirectional-causation reading of Stage 6 (A = 26%, S = 49%, A+S = 25%). Conditional implications follow for Qur’anic translation and Islamic psychology; small per-root n and the closed-text status of the corpus bound the inferences.

Keywords: Qur’an; anger spectrum; semantic field; cognitive linguistics; scalar semantics; conceptual metaphor; corpus linguistics.

1. Introduction

1.1. Statement of the problem

The Qur’anic lexicon for the anger spectrum exhibits a precision that survives only with difficulty into translation: where English versions level *uff*, *karh*, *ḍīq*, *ḥuzn*, *asaf*, *naqm*, *sakhaṭ*, *maqt*, *ghaḍab*, *ḥard*, *ghayz*, *baghy*, *ṭughyān*, and *‘utuww* into a small set of broad equivalents — “displeasure,” “distress,” “grief,” “anger,” “rage,” “transgression,” “rebellion” — the Arabic original distinguishes them with the precision of a graded taxonomy. We argue that these fourteen roots are not synonyms but coordinated nodes in a single graded semantic field organised along intensity of action, traversing pre-anger displeasure, inner pressure and contraction, evaluative aversion, active anger, compressed/explosive rage, and behavioural outcomes (*baghy*, *ṭughyān*, *‘utuww*) whose causation is bidirectional rather than purely emotional. The distance from *uff* (*sūrat al-Isrā*, Q. 17:23) to *‘utuww* (*sūrat al-Furqān*, Q. 25:21) is graded, not categorical.

1.2. Aims and originality

This paper combines Izutsu’s semantic-field approach, Kennedy–Sassoon scalar semantics, and Lakoff–Kövecses CMT to produce stronger philological claims than any framework in isolation. Izutsu’s method yields a network without an internal ordering; Lakoff–Kövecses supply an English-anchored cognitive schema, and the two-volume *Metaphors of ANGER across Languages* (Kövecses, Benczes and Szelid 2025) omits Classical Arabic; Kennedy–Sassoon orders predicates along a dimension but has not been applied to a Qur’anic emotion field. Qā’imīniyā (2011, 2014) and Pakatchi and Afrashi (2020) already integrate Izutsu with CMT in Persian; the novel element here is the scalar layer operationalised as a six-stage continuum and tested against a 312-attestation reproducible pipeline, alongside (i) a $\kappa = 0.79$ -validated tafsīr-coding audit of the bidirectional-causation reading of Stage 6, (ii) the *kaḥm-as-tying-off-of-waterskin* analysis of agent-internal containment, and (iii) the framing of Q. 67:8 of *sūrat al-Mulk* as an early and unusually transparent exemplification of vessel-internal failure, calibrated against the six-tradition survey of §5.5 acknowledging Job 32:19 (*bāqa’*) as a structurally cognate Northwest-Semitic precedent. The interpretive payoff: the continuum constitutes a coherent logic of escalation with structural affinities to modern emotion psychology, exceeding the contemporary models at Stage 6, where *baghy/ṭughyān/‘utuww* extend the regulatory horizon of Gross’s process model into the moral-anthropological register.

1.3. Prior work and the gap

Prior scholarship clusters into five lines. Single-word lexical studies (Narimani, Iqbali and Chari 2021 on *ghayz/ghaḍab/sakhaṭ*; Qarizadeh et al. 2024 on *ṭughyān*) supply high-quality per-lexeme analysis but no integrated continuum. Semantic-field and cognitive-semantic studies in the Izutsu tradition (Qā’imīniyā 2011,

2014; Pakatchi and Afrashi 2020; Albayrak 2020) leave the graded-continuum layer untreated. Ethics-of-anger work in the *kazm al-ghayz* tradition is normatively rich and linguistically thin. Cross-linguistic CMT literature, updated by Kövecses, Benczes and Szelid (2025) across twenty-two languages, omits Classical Arabic; the closest Arabic-CMT precedents (Maalej 2004, 2007; Rabab'ah and Al-Saidat 2022) treat contemporary dialects. Historical-critical and corpus-linguistic work supplies complementary apparatus (Sinai 2017, 2023; Neuwirth 2019; El-Awa 2006; Sharaf and Atwell 2012a, 2012b; Akra, Hammouda and Jarrar 2025; Sawalha et al. 2024); translation-criticism is now quantitative at scale (Gaanoun and Alsuhaibani 2025; Khan et al. 2025) but lacks a graded lexeme-level reference model. The present study occupies the unfilled intersection: it adds the Kennedy–Sassoon scalar layer to the Izutsu/CMT stack, applies it to the anger field the 2025 Metaphors of ANGER encyclopedia leaves un-canvased, and releases the pipeline under Sharaf–Atwell / QuranMorph reproducibility standards.

1.4. Research questions and hypotheses

We address five research questions: (RQ₁) the internal structure of the anger-spectrum field and its principal nodes; (RQ₂) the action-intensity ordering and its theoretical legitimation; (RQ₃) corpus-distributional corroboration (frequency, Meccan/Medinan, morphology, network); (RQ₄) structural homology with Gross, Plutchik, Spielberger, and Lazarus–Scherer appraisal theory; (RQ₅) implications for Qur'anic translation and Islamic psychology.

The principal Qur'anic Action-Intensity Continuum Hypothesis holds that the fourteen roots operate as graded steps on a single semantic continuum structured by intensity of action, organised in six phenomenological stages. Five subsidiary hypotheses follow: (H₁) non-uniform six-stage frequency distribution; (H₂) near-equal split between phenomenological core (S₁–5) and behavioural pole (S₆), consistent with the bidirectional Stage-6 reading; (H₃) discursively-specialised exclusive-Medinan distribution of *sakhat*; (H₄) *baghy* as bridge node in the co-occurrence network; (H₅) Q. 67:8 of *sūrat al-Mulk* as an unusually transparent exemplification of the vessel-internal failure construal.

2. Theoretical Framework

2.1. Izutsu's semantic-field method

Izutsu (1959, 1964, 2002) argued that the central Qur'anic terms are nodes in a network whose aggregate constitutes a Qur'anic *Weltanschauung*. His three-layer apparatus (focus words, semantic fields, key words) has been productive in Persian-language Qur'anic semantics (Qā'imīniyā 2011, 2014; Pakatchi and Afrashi 2020). The fourteen target roots constitute the field around the focus words *ghaḍab* (central anger) and *baghy* (central behavioural-outcome). Izutsu's method is silent on the question that drives the present enterprise: granted that these lexemes co-belong to a field, what is the structural relation between them within it? For this we require scalar semantics.

2.2. Scalar semantics: from network to continuum

Gradable-predicate theory (Kennedy 1999, 2007; Sassoon 2010) supplies the missing apparatus. A scale has three components — a dimension of measurement, an ordering, and a unit of comparison. The Qur'anic anger field is such a scale: dimension = intensity of action (decomposable into locus, experiential intensity,

scope of consequence); ordering = the six-stage progression; implicit unit = degree of externalisation from inner phenomenology to social rupture. Without this framework, the claim that *ghayẓ* is “more intense than” *ghaḍāb* relies on tacit native-speaker intuition. Kennedy–Sassoon was developed for gradable adjectives; its extension here assumes a category-neutral underlying intensity scale recoverable from lexicographic and corpus-distributional evidence.

2.3. Conceptual metaphor theory and the container schema

Conceptual metaphor theory (Lakoff and Johnson 1980; Lakoff and Kövecses 1987; Kövecses 2000, 2010) treats metaphor as a cognitive structure mapping concrete source domains (container, heat) to abstract targets (emotion). Lakoff and Kövecses (1987: 195–221) reconstruct American-English anger as a five-stage prototype scenario — (1) offending event, (2) anger, (3) attempt at control, (4) loss of control, (5) retribution — onto which ANGER IS THE HEAT OF A FLUID IN A PRESSURIZED CONTAINER, ANGER IS FIRE, and ANGER IS A WILD ANIMAL are mapped, with the container/pressure cluster operating especially in stages (2)–(4).

2.3.1. Primary vs. complex metaphors and the source–target mapping Following Grady (1997) and Kövecses (2020), primary metaphors are grounded directly in recurrent embodied experience (ANGER IS HEAT; EMOTIONAL CONTAINMENT IS PHYSICAL CONTAINMENT), while complex metaphors compose primary metaphors with culturally-elaborated folk physiology. PRESSURIZED-CONTAINER ANGER is a complex metaphor — the body construed as a sealed vessel, anger as a hot fluid inside it, loss of control as the fluid’s exit or the vessel’s failure. The Qur’anic *ghayẓ*–*kāẓm*–*tamayyuz* triad operationalises this complex construct, not the primary ANGER IS HEAT schema alone. Source-to-target mappings: HOT FLUID under pressure → INTENSE ANGER (*ghayẓ*); CONTAINER (sealed vessel) → BODY/SELF (chest, *qalb*, *naḥs*); SEALING / TYING-OFF the vessel (*kāẓm* as waterskin closure; Ibn Manẓūr, *Lisān* 12:519, s.v. *k-ẓ-m*) → AGENT-INTERNAL SUPPRESSION; SEPARATION OF CONSTITUENTS within the vessel → VESSEL-INTERNAL FAILURE MODE (the *tamayyuz* construal, §4.7.2). The triad maps tightly onto stages (2)–(4) of the Lakoff–Kövecses scenario, with Q. 67:8 providing an unusually transparent linguistic crystallisation of stage (4).

2.3.2. Cultural Linguistics: universal schema, Qur’anic elaboration Sharifian’s (2011) Cultural Linguistics distinguishes what is universal in CMT from what is culturally elaborated. The primary metaphors are cross-linguistically robust and find support in the Qur’anic data, consistent with CMT’s universalist claim. The culturally elaborated layer — the Qur’an’s distribution of affect across *qalb*, *ṣadr*, and *naḥs* as differentiated loci, the moral commendation of *kāẓm* in the *taqwā* register, and the *tamayyuz* construal of Q. 67:8 — is what the Qur’anic data illuminate beyond the universalist baseline (returned to in §5.4).

2.4. The integrated framework

Izutsu identifies field boundaries; scalar semantics orders members along the action-intensity dimension; CMT grounds the pressurisation-collapse complex metaphor organising Stage 5.

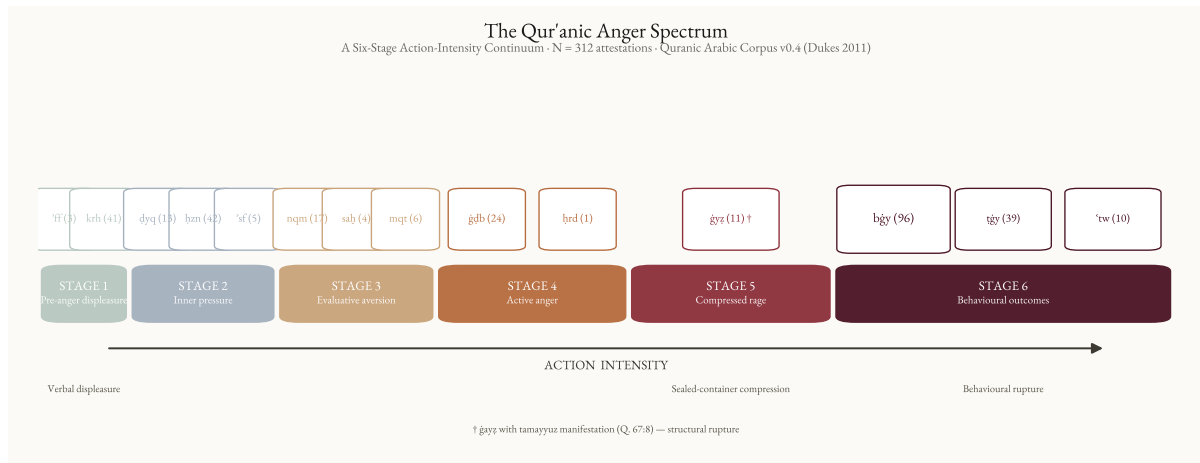


Figure 1. Figure 1. The six-stage Qur'anic anger-spectrum continuum along the action-intensity axis; node area scales with QAC attestation frequency. S1 pre-anger displeasure ('ff–krh); S2 inner pressure (dyq–hzn–'sf); S3 evaluative aversion (nqm–sxt–mqt); S4 active anger (ghdb–hrd); S5 compressed/explosive rage (ghy; tamayyuz manifestation); S6 behavioural outcomes (bghy–tghy–'tw) with bidirectional causation (§4.8.5).

3. Methods

3.1. Corpus

The empirical analysis uses the Quranic Arabic Corpus (QAC) v0.4 (Dukes 2011, GPL): a morphologically tagged record for each of the 128,219 segmentable units, specifying location, surface form (Buckwalter), POS, and morphological features. Verse text in Uthmanic rasm and Meccan/Medinan sura classification are drawn from the Tanzil project (2008, CC-BY-ND 3.0).

3.2. Selection of target roots

The fourteen core roots were selected by three concurrent criteria: (i) exegetical centrality — focal reference in at least three of the four authoritative lexica (Maqāyīs, Mufradāt, Lisān, al-Mostafawī's Taḥqīq); (ii) minimum corpus frequency of one attestation, admitting hapax legomena (ḥard, Q. 68:25) when supported by three of the four lexica and consistent tafsīr treatment; (iii) thematic membership without significant overlap with adjacent fields. Excluded candidates (transparency): q-h-r, ḥ-n-q (not attested), 'n-f, d-r-b, q-ṣ-w/q-s-y, gh-l-z, 'n-f, ṣ-n-. Karh is borderline — karāha affective, ikrāh structural — included on the karāha sense (§4.3.1). Five umbrella moral terms (šn', jrm, fsq, zlm, 'dw) are used for robustness (§4.9) and cross-field co-occurrence (Table 4); excluded from the principal continuum because of inflated cardinality (zlm n = 315 functions as a meta-category) or adjacency to normative-ethics fields.

3.3. Computational pipeline

The Python (MIT-licensed) pipeline parses the QAC morphology file, filters on target roots, collapses multi-segment morphemes to surface words, joins sura-level Meccan/Medinan metadata, and emits master and per-root concordances, distributional summaries, and network/centrality CSVs; the complete analysis is reproducible from a fresh clone in four commands.¹

¹QAC v0.4 is pinned by SHA-256; outputs are released under MIT (code) and CC-BY 4.0 (manuscript/figures). The fourteen exemplar verses plus a stratified fifty-segment audit on the four most frequent core roots (hzn, bghy, ghdb, tghy) returned 100% correct segment-to-root assignment against Tanzil; full audit at [data/concordance/validation_audit.md](#).

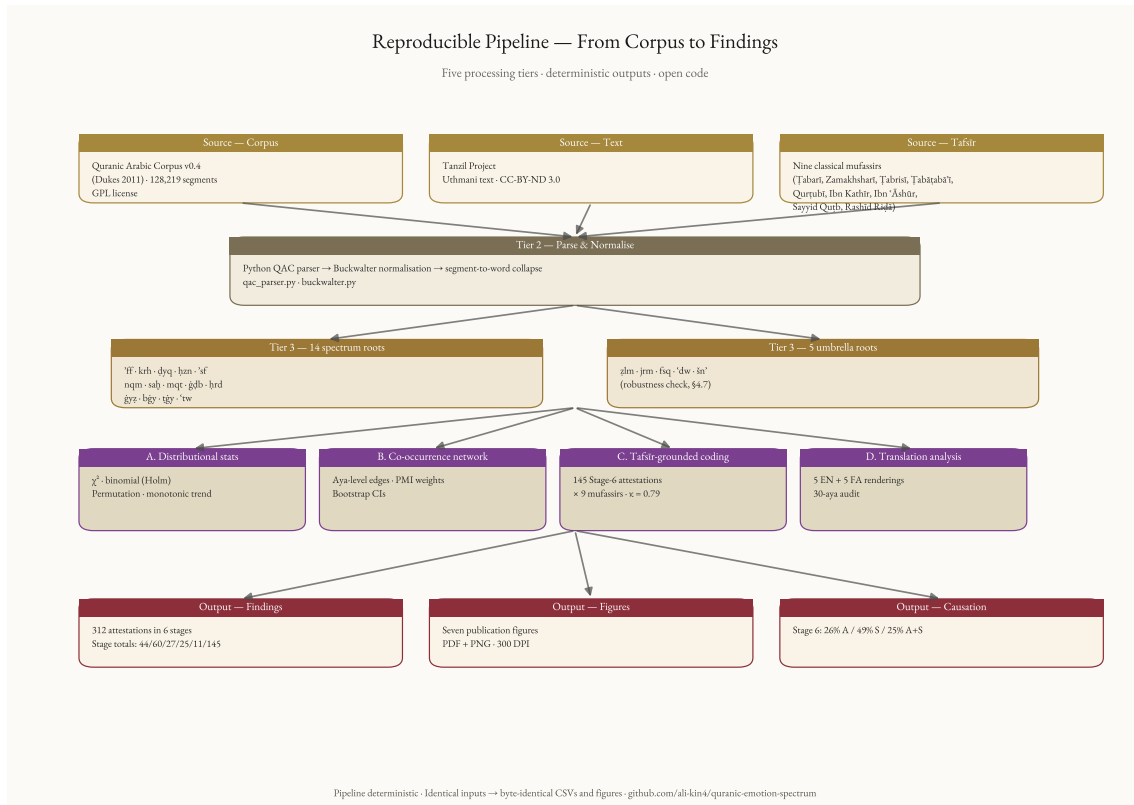


Figure 2. Figure 8. End-to-end pipeline from corpus to findings: QAC vo.4 and Tanzil sources → Buckwalter→Unicode parse → fourteen spectrum + five umbrella-moral root selection → distributional, network, tafsir-coding, and translation-audit analyses → 312 attestations, figures, and Stage-6 causation breakdown. The pipeline is deterministic.

3.4. Quantitative analyses

Quantitative tests beyond frequency tabulation: (a) $\chi^2(5)$ and $\chi^2(1)$ on uniform-stage and S1–5/S6 nulls, with a marginal-preserving permutation null (Cochran 1954; Patefield 1981; Mehta and Patel 1983; Agresti 2013, §3.5);² (b) three monotonic-trend tests (Spearman ρ , Mann–Kendall, Cochran–Armitage-style OLS slope); (c) two-sided exact binomials against the 0.74 Meccan-ayat baseline, Holm–Bonferroni-adjusted over the family of fourteen per-root tests, with sensitivity tables at baselines 0.70–0.78 and a family-size sensitivity check;³ (d) morphological breakdown by POS; (e) aya-level co-occurrence network (cf. Mottaghi et al. 2024) with weighted degree, betweenness, closeness, and eigenvector centrality, bootstrap 95% percentile CIs (10,000 ayat-resamples; fixed RNG seed).

3.5. Qualitative analyses

Qualitative complements: (a) comparative etymology across the four lexica (Table 1); (b) comparative tafsīr on nine commentaries — al-Ṭabāṭabāʾī, al-Zamakhsharī, al-Ṭabarsī, Ibn ʿĀshūr, al-Ṭabarī, al-Qurṭubī, Ibn Kathīr, Sayyid Quṭb, ʿAbduh–Riḍā, with al-Qushayrī’s *Laṭāʾif al-Ishārāt* for Sufi cross-checking on *kaẓm al-ghayẓ*; (c) phenomenological analysis of contextual markers (*ḍiq al-ṣadr*, *ḥuẓn al-qalb*, *ḍāqa dharʿan*); (d) illustrative translation-criticism on five English translations at canonical exemplar verses (§4.10).

3.6. Validity check

Fourteen headline exemplar verses — one per core root — were independently verified against the corpus output (Q. 17:23, 2:216, 15:97, 2:38, 43:55, 7:126, 47:28, 40:10, 48:6, 68:25, 3:134 [with Q. 67:8 *tamayyuz*], 42:42, 96:6, 25:21); all matched the Uthmani edition.

4. Findings

4.1. The four-lexicon etymology table

Table 1 summarises the canonical glosses of each root across four reference lexica.

Table 1. Comparative etymology of the fourteen core roots across the four authoritative Arabic lexica

²The permutation null holds per-root counts fixed and randomly permutes the fourteen stage labels (10,000 permutations, fixed RNG seed 20260509); the empirical p is the fraction of permuted χ^2 values meeting or exceeding 227.15. This is the natural extension of Fisher-exact reasoning to a many-cell design (Patefield 1981; Mehta and Patel 1983).

³The reported family size of fourteen is the per-root Meccan/Medinan binomial family. A strict reading would expand the family to ≈ 20 (adding the omnibus $\chi^2(5)$, S1–5/S6 $\chi^2(1)$, three trend tests, and A-only re-analysis). Under the strict family, *karh* and *baghy* remain below $\alpha = 0.05$; *ghayẓ* moves from adj. ≈ 0.019 to 0.027 (still significant); *sakhaṭ* moves from adj. 0.050 to 0.075 (substantively informative — see §4.5.2). Conclusions of §4.5–§4.8 are robust to the family-size choice.

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-‘Arab (Ibn Manẓūr)	Taḥqīq (al-Mostafawi)
I	’-f-f	“kalimatu taḍajjurin” (a word of vexation); smallest audible expression of displeasure	the lowest verbal expression of displeasure (adnā kalimati al-taḍajjur)	term of irritation (taḍajjur) directed at someone or something disliked; cf. uffah (sub-vocal sigh)	minimal vocal sign of inner aversion; pre-anger affective register
I	k-r-h	dislike, finding burdensome; coercion (ikrāh)	aversion of the soul; structurally extended to coercion	what is disliked by nature or by reason	inner karāha vs. structural ikrāh (the latter is not an emotion)
2	ḍ-y-q	constriction; absence of opening	the state of being pressed upon, contrasted with capacity (sa‘ah)	reduction of available space, from sa‘ah to ḍīq	inner or outer absence of expansion (bast)
2	ḥ-z-n	rough/uneven ground → heaviness of soul	sorrow as the opposite of joy (farah)	rugged ground, then roughness of the self	affective response to perceived loss
2	’-s-f	intense grief mixed with anger	extreme anger conjoined with grief	reaching the limit of grief and anger	composite of grief and proximate anger
3	n-q-m	al-naqimah: disapproval (inkār) coupled with intent to punish; cf. intaqama min = exact retribution from	objecting to (ankara ‘alā) and seeking redress (‘uqūbah) against a reprehensible act	strong denunciation (ankara ashadda al-inkār) yoked to ‘uqūbah; basis of divine attribute al-Muntaqim	evaluative anger oriented toward retribution; cognition-grounded, action-tilted

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-‘Arab (Ibn Manẓūr)	Taḥqīq (al-Mostafawi)
3	s-kh-ṭ	severe aversion and ethical rejection	the opposite of riḍā (consent), with intensity	severe non-acceptance with judgment	declarative rejection grounded in moral assessment
3	m-q-t	the most intense form of ibghāḍ (Zajjāj)	aversion arising from observing a qabiḥ act (al-Rāghib)	ashaddu al-ibghāḍ; aversion-distance rather than approach-aggression	purely evaluative moral aversion
4	gh-ḍ-b	inner agitation and motion	the boiling of the heart’s blood seeking retribution	a rising perturbation of the soul	directed emotion enabling retributive action
4	ḥ-r-d	ḥarada: to be angry; also the act of qaṣd (deliberate aim)	conjoining ghaḍab with qaṣd: anger directed by purpose, viz. denial	ḥarada “to act/move toward an aim with resolve while in anger”; cf. ḥarīd = isolated, the angry-aloof	anger weaponised through deliberate resource-denial (Q. 68:25 aṣḥāb al-jannah); affective + volitional + miserly composite
5	gh-y-z	pressure and surge, contained	the most intense forms of contained anger	filling of the self with restrained anger	high-pressure emotional accumulation
6	b-gh-y	seeking beyond rightful bounds	overstepping the path of right	dominance through aggression	aggressive action with intent to oppress
6	ṭ-gh-y	overflow of water past its bank	crossing the permitted measure	severe transgression of the limit	rebellious shattering of normative bounds

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-‘Arab (Ibn Manẓūr)	Taḥqīq (al-Mostafawi)
6	‘-t-w	severe rebellion with arrogance	extreme self-aggrandisement in disobedience	transgressing the limit, with istikbār-attachment as collocational feature	entrenched rebellious posture (a property of moral character)

The table grounds two distinct claims at different evidential strengths. The pairwise ordering relations among many root-pairs (e.g. uff < ghaḍab; karāha < maqt; ghaḍab < ghayz; naqm < baghy) are internally encoded in the lexicographic tradition: the four authoritative lexica already gloss each root with progressively externalised and structural meanings. The partition into six discrete phenomenological stages, by contrast, is the analytical contribution of the present study, licensed jointly by the lexicographic ordering, the tafsīr readings of §§4.3–4.8, and the morphological and distributional evidence of §4.2.

4.2. Distributional summary

Table 2 and Figures 2–5 report the principal distributional facts for the fourteen roots: per-root and per-stage frequency, Meccan/Medinan split, and morphological breakdown by POS.

Table 2. Distributional profile of the fourteen core roots (Wilson-score 95% CIs on the Meccan-rate point estimate; baseline = 0.74)

Stage	Root	n	M	Md	M-rate (95% CI)	Notes
1	’ff	3	3	0	1.00 [0.44, 1.00]	All Meccan; n too small for inference
1	krh	41	14	27	0.34 [0.22, 0.49]	Medinan-tilted, CI excludes baseline; Holm-adj. $\approx 1.4 \times 10^{-6}$
2	dyq	13	9	4	0.69 [0.42, 0.87]	CI includes baseline; inner constriction

Stage	Root	n	M	Md	M-rate (95% CI)	Notes
2	ḥzn	42	26	16	0.62 [0.47, 0.75]	CI includes baseline; valence transversal (\$4.4.2)
2	'sf	5	5	0	1.00 [0.57, 1.00]	Underpowered; CI excludes baseline at boundary only
3	nqm	17	12	5	0.71 [0.47, 0.87]	CI includes baseline; not FDR- significant
3	sxt	4	0	4	0.00 [0.00, 0.49]	CI excludes baseline; Holm-adj. = 0.050
3	mq̣t	6	4	2	0.67 [0.30, 0.90]	Wide CI; includes baseline
4	ghḍb	24	13	11	0.54 [0.35, 0.72]	CI excludes 0.74 baseline; central anger lexeme
4	ḥrd	1	1	0	—	Hapax: sūrat al-Qalam, Q. 68:25
5	ghyz	11	3	8	0.27 [0.10, 0.57]	Medinan- tilted; Holm-adj. $\approx$ 1.9×10^{-2} — survives FDR

Stage	Root	n	M	Md	M-rate (95% CI)	Notes
6	bghy	96	55	41	0.57 [0.47, 0.67]	CI excludes baseline; Holm-adj. $\approx$ 5.3×10^{-3} — survives FDR; bridge node (§4.8.5)
6	tghy	39	29	10	0.74 [0.59, 0.85]	CI includes baseline; not FDR- significant
6	‘tw	10	9	1	0.90 [0.60, 0.98]	CI includes baseline; istikbār- attached
Σ	—	312	183	129	—	

We lead the small-n roots (asaf at $n = 5$, sakhaṭ at $n = 4$, mqt at $n = 6$) with the Wilson-score CI rather than the p-value; the Holm-Bonferroni adjustment then carries the inferential weight across the family.

Across the six stages: $S_1 = 44$; $S_2 = 60$; $S_3 = 27$; $S_4 = 25$; $S_5 = 11$; $S_6 = 145$. Two distinct nulls bear on these data, answering different questions. The uniform-six-stage null (is the per-stage distribution uniform?) is unambiguously rejected by the asymptotic $\chi^2(5) = 227.15$, $p < 0.001$ (Cramér’s $V = 0.38$, Cohen’s $w = 0.85$), since baghy alone supplies 96 of 312 attestations. A marginal-preserving permutation null holding per-root counts fixed and randomly permuting the fourteen stage labels tests instead “conditional on baghy’s mass-dominance, is our partition more skewed than a random six-binning of the same roots?” — the empirical $p = 0.24$ says no. This is not evidence that the distribution is uniform (already ruled out); it is evidence that the omnibus χ^2 cannot discriminate our partition from alternative six-bin partitions of the same roots. The structural case for the six-stage architecture rests on the qualitative evidence of §4.1 and §4.3–§4.8 together with the morphological-gradient and Meccan/Medinan-specialisation evidence below.

The S_1 – S_6 $\chi^2(1) = 1.55$ (Cohen’s $w = 0.07$) fails to reject equal split between the phenomenological core (167) and the behavioural pole (145) — absence of evidence of a difference, consistent with the bidirectional-causation reading of §4.8.5 and §5.3.

4.2.1. Monotonic-trend tests Three monotonic-trend tests of stage rank against attestation frequency all fail to reject the no-trend null: Spearman $\rho = -0.09$ ($p = 0.85$), Mann–Kendall $z = -0.38$ ($p = 0.71$), Cochran–Armitage-style OLS slope = 10.17 ($p = 0.40$). The continuum orders by semantic intensity, not attestation frequency. The Qur’an is a closed hortatory text whose lexical weights are pedagogically determined: density is heaviest at the two poles of moral consequence (S_1 –2, entry points for *kazm*, *taqwā*, *ḥilm*; S_6 , the structural

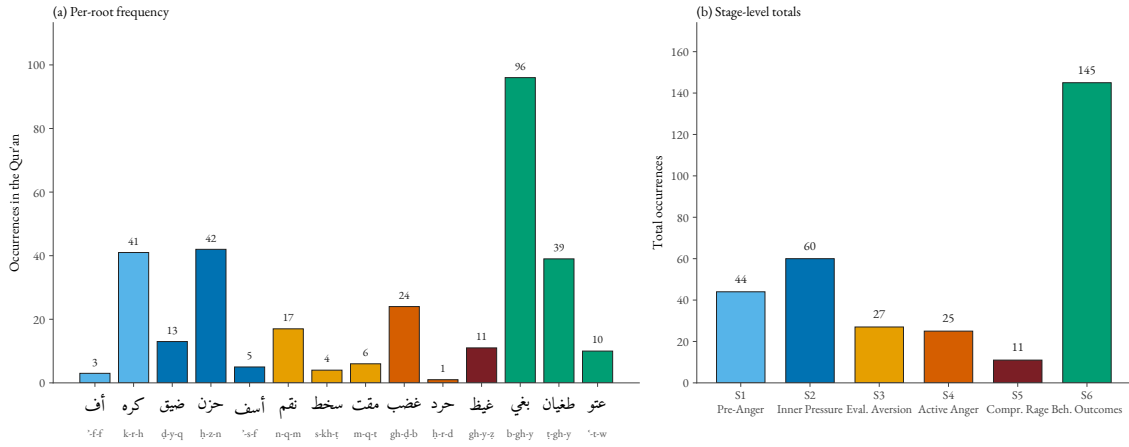


Figure 3. Figure 2. Per-root and per-stage frequency of the 312 core attestations. Left: per-root frequency (bar colour encodes stage). Right: stage totals $S_1 = 44$, $S_2 = 60$, $S_3 = 27$, $S_4 = 25$, $S_5 = 11$, $S_6 = 145$. $\chi^2(5) = 227.15$ asymptotic, but permutation null $p = 0.24$ (§4.2); $S_1\text{--}5/S_6 \chi^2(1) = 1.55$, consistent with the bidirectional-causation reading of §4.8.5.

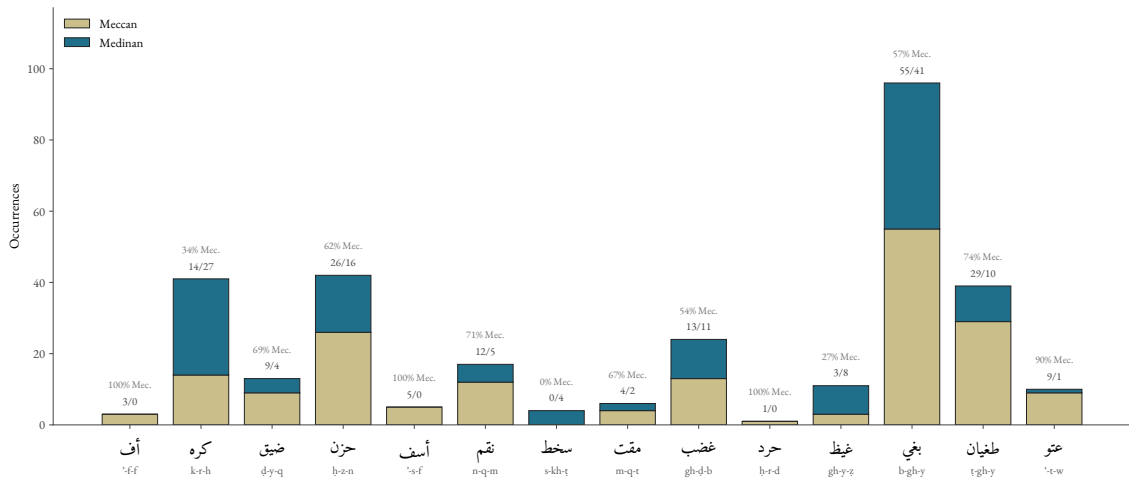


Figure 4. Figure 3. Stacked Meccan/Medinan distribution. Three roots survive Holm-Bonferroni correction ($\alpha = 0.05$): karh (Medinan), baghy (Meccan), and ghayṣ (Medinan); sakhaṭ at adj. $p = 0.050$. Asaf, naqm, ‘utuww tilts are contextual. The patterns instantiate discursive-contextual specialisation (§5.4).

pathologies most frequently denounced), with $S_3\text{--}5$ as middle-trajectory states. A flat trend is what the theory predicts.

4.3. Stage 1 — Pre-anger displeasure

4.3.1. Uff (3; all Meccan). Uff (taḍajjur) is the Qur’anic register’s lowest verbal expression of displeasure. Its three attestations occupy paradigmatic moral positions: sūrat al-Isrā’ (Q. 17:23) prohibits saying uff to one’s parents — the lowest sign of displeasure weighted at the highest relational stake; sūrat al-Anbiyā’ (Q. 21:67) places it in Abraham’s mouth confronting idol-worshippers; sūrat al-Aḥqāf (Q. 46:17) ascribes it to a rebellious child reproaching believing parents.

4.3.2. Karh (41; 14M/27Md; Holm-adj. $p \approx 1.4 \times 10^{-6}$ — survives FDR). Karh exhibits a twofold structure: inner karāha (pre-anger evaluative dislike, framed as reformable by sūrat al-Baqara, Q. 2:216, ‘asā an takrahū shay’an wa-huwa khayrun lakum; the S_1 sense), and structural ikrāh (coercion — not an emotion but a structural extension into compulsion: Q. 2:256, lā ikrāha fī al-dīn; Q. 16:106; Q. 24:33). Inclusion rests on

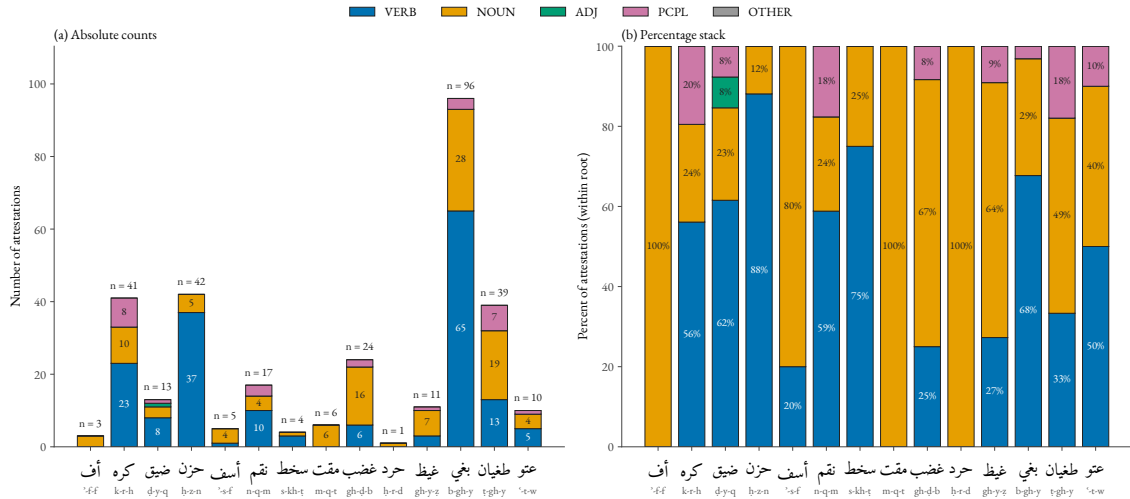


Figure 5. Figure 5. Morphological profile by POS. Verbal dominance at S1–2 (hzn 88% verbal, đyq 62%), nominalisation at S4 (ghđb 16n/6v), peak nominality with participial forms at S6 (ṭāghūt, bāghin). The verbal→nominal gradient is independent corpus-internal evidence for the action-intensity continuum.

the karāha sense; the structural ikrāh in Medinan community-discipline contexts (consent in marriage, freedom in religion) accounts for the Medinan tilt.

4.4. Stage 2 — Inner pressure and contraction

Stage 2 brings together three roots (đ-y-q, ḥ-z-n, ’-s-f) with sixty attestations.

4.4.1. Đīq (13; 9M/4Md). Ibn Fāris glosses đyq as constriction/narrowness; al-Rāghib opposes it to sa’ah. The root appears predominantly in đīq al-ṣadr (constriction of the chest), localising the experience to the embodied site (sūrat al-Ḥijr, Q. 15:97, the prophetic response to social rejection). Al-Ṭabāṭabā’ī (al-Mīzān 12:195, ad Q. 15:97: wa-laqaḍ na’lamu annaka yađīqu ṣadruka bimā yaqūlūn) reads the verse as expressing divine knowledge of prophetic đīq rather than prohibition. The compound ḍāqa bihim dhar’an in the Lot narrative (sūrat Hūd, Q. 11:77/80) figures post-anger blocked aggression: trigger sī’a bihim → constriction → aspirational voicing law anna lī bikum quwwatan → transfer of agency to the divine. Đyq thus sits at both poles of the spectrum (pre-anger contraction and upper-pressure inhibition); the morphological profile (8v / 3n / 1ptcp) registers a transient, situationally-induced state.

4.4.2. Ḥuzn (42; 26M/16Md). The most frequent and most verbal S2 lexeme (88% verbal). Ibn Manẓūr preserves the primary sense “rugged, uneven ground”; the metaphorical extension figures grief as inner roughness (anticipating Kövecses 2000 on EMOTION IS A LANDSCAPE). Sūrat al-Baqara (Q. 2:38) pairs ḥuzn with khawf and opposes both to divinely-given guidance. Ḥuzn has four valences: standalone grief over loss (Jacob over Joseph, sūrat Yūsuf, Q. 12:84), externally-induced sorrow from being wronged, positive moral compunction (the affective register of tawba), and anger turned inward when externally inhibited; inclusion at S2 is justified by corpus density and proximity of the second and fourth valences to the anger field.

4.4.3. Asaf (5; 5M/0Md). Al-Rāghib glosses asaf as the conjunction of grief and anger (al-jam’ bayna al-ḥuzn wa-l-ghaḍab). Sūrat al-Zukhruf (Q. 43:55, fa-lammā āsafūnā intaqamnā minhum) frames it as the immediate causal precursor of divine retribution (al-Ṭabarsī, Majma’ 9:82). The compound ghaḍbān asifan, applied to Moses on his return to the calf-worshippers (Q. 7:150; Q. 20:86), generates a three-layer dialectic in which ghaḍab is outward-kinetic and asaf the inward-compassionate ground. The 5/0 distribution is power-

bound (Wilson CI [0.57, 1.00]). The aya-level network (Figure 4) confirms S2 coherence: ḥzn-ḍyq is the strongest single edge (weight 3).

4.5. Stage 3 — Evaluative aversion

Stage 3 brings together three roots (n-q-m, s-kh-ṭ, m-q-t) with twenty-seven attestations marking distinct modes of evaluative aversion. Maqt is the lexicographic superlative of aversion (Zajjāj, ashaddu al-ibghāḍ) but is placed between sakhaṭ and ghaḍab on the action-intensity axis because the dimension here is intensity of action, not intensity of aversion.

4.5.1. Naqm (17; 12M/5Md). Naqm is vengeful disapproval — anger grounded in moral evaluation and oriented toward retribution. Sūrat al-Aʿrāf (Q. 7:126, wa-mā naqamū minnā illā an āmannā bi-āyāti rabbīnā) is the linguistic exemplar; al-Ṭabarī (Jāmiʿ al-bayān 13:23 ad loc.) glosses naqamū as ʿābū wa-ankarū, exposing the taʾaṣṣub-structure of disbelief (al-Qurṭubī, al-Jāmiʿ 7:262 convergent). The lexeme generates al-muntaqim as a divine attribute (Q. 32:22 and parallels). The Wilson CI includes the baseline.

4.5.2. Sakhaṭ (4; 0M/4Md). The exclusive-Medinan distribution is one of the paper’s principal distributional findings.⁴ Holm-adjusted $p = 0.050$ — borderline. All four attestations are tied to the Munāfiqūn in their Medinan-period exegesis: sūrat Muḥammad (Q. 47:28, ittabaʿū mā askhaṭa Allāh wa-karihū riḍwānahu) is canonical, though al-Ṭabarī notes that sūrat al-Māʾida (Q. 5:80) is exegetically discussed as directed at the People of the Book rather than the Munāfiqūn properly. Following Narimani, Iqbali and Chari (2021), we read sakhaṭ as sustained ethical disapprobation rather than instantaneous emotional surge; we propose:

Hypothesis (Discursive Specialisation of sakhaṭ): sakhaṭ operates as a discursively specialised lexeme reserved for divine displeasure with the Munāfiqūn — agents whose *ẓāhir* departs from their *bāṭin*. Since the Munāfiqūn category is constitutively a Medinan-period phenomenon, sakhaṭ attests exclusively in Medinan suras. This is one instance of a general phenomenon of discursive-contextual lexical specialisation, of which *karh* (Medinan-community discipline) and the umbrella patterns *jrm* (Meccan) / *fsq* (Medinan) are convergent.

4.5.3. Maqt (6; 4M/2Md). Maqt is intense moral aversion: the four lexica converge on ashaddu al-ibghāḍ (Zajjāj); al-Rāghib grounds it in observation of a reprehensible act, making maqt purely evaluative. The canonical exemplar is sūrat Ghāfir (Q. 40:10, la-maqtu Allāhi akbaru min maqtikum anfusakum), the doubled maqt comparing divine moral aversion at the disbelievers with their own retrospective self-maqt on the Day of Resurrection (al-Ṭabarī, Jāmiʿ al-bayān 21:351 ad loc.; al-Qurṭubī, al-Jāmiʿ 15:296 convergent). Sūrat al-Nisāʾ (Q. 4:22), sūrat Fāṭir (Q. 35:39), and sūrat al-Ṣaffʿ (Q. 61:3) stabilise the lexeme as moral aversion as evaluative response, distinct from the kinetic-motor ghaḍab of Stage 4.

4.6. Stage 4 — Active anger

4.6.1. Ghaḍab (24; 13M/11Md). The central anger lexeme — near-balanced Meccan/Medinan distribution marks it as the unmarked default. The four lexica gloss it as inner agitation oriented toward retribution (al-Rāghib) or inner motion disposing the agent to action (Ibn Fāris); predicated balance-fully of God (sūrat al-Fath, Q. 48:6), prophets, and believers. Morphologically 67% nominal (16n/6v/2 ptcp) vs ḥuzn’s 88% verbal:

⁴The 0.74 baseline is the proportion of ayat (not suras) classified Meccan in the QAC metadata; we use the ayat-weighted rate because an attestation drawn at random from the corpus hits an aya, not a sura. A sensitivity table over baselines 0.70–0.78 yields two-sided p values from 0.0081 (at 0.70) to 0.0023 (at 0.78); the conclusion is robust across the range.

ghaḍab is a structural concept-state, not a transient affect. Al-Rāzī (Mafātīḥ al-Ghayb) glosses divine ghaḍab in the register of irādat al-‘uqūbah — a normative-juridical orientation, not an emotional response. Network analysis places ghḍb among the top three for betweenness (point estimate 0.17).

4.6.2. Ḥard (Q. 68:25, hapax). A hapax in the Garden-Owners parable (wa-ghadaw ‘alā ḥardin qādirīn): owners vowing in the night to harvest before dawn so as to deny the poor their share. Al-Ṭabarī (Jāmi‘ al-bayān 23:550 ad loc.) records four classical qawls: (i) qaṣd (purposive aim — Mujāhid, Ibn Zayd, Ibn ‘Abbās ‘an ‘Ikrima); (ii) ghaḍab (anger toward the poor — Qatāda, al-Suddī); (iii) man‘ (withholding — al-Ḥasan); (iv) quwwa/sū‘ (strength or evil intent — Sufyān al-Thawrī). The affective synthesis — ḥard as anger weaponised through deliberate denial — converges (i) and (ii), defended by al-Ṭabrisī (Majma‘ 10:90) and al-Ṭabāṭabā‘ī (al-Mīzān 19:380); it is not the unanimous consensus but the reading we adopt, because the narrative consequence (divine destruction of the garden) requires both an affective and an intentional component.

4.7. Stage 5 — Compressed/explosive rage

4.7.1. Ghayḥ and the kaẓm metaphor (11; 3M/8Md; Holm-adj. $\approx 1.9 \times 10^{-2}$ — survives FDR). The four lexa converge: ghayḥ is compressed contained anger (al-Rāghib), the highest concentration of restrained anger (al-Mostafawi), filling of the self with restrained anger (Ibn Manẓūr) — pressurised contents of a sealed vessel. Sūrat Āl ‘Imrān (Q. 3:134) commends al-kāẓimīn al-ghayḥ; kaẓm derives from the tying-off of a waterskin (Ibn Manẓūr, Lisān 12:519, s.v. k-ẓ-m), the agent-internal regulation that sustains containment. The construction itself presupposes that the believers being commended experience the ghayḥ they suppress; the Narimani et al. claim that ghayḥ in human reference is exclusive to unbelievers must therefore be qualified — transitive ghayḥ directed at believers is exclusively non-believer-attributed, but intransitive/experiential ghayḥ (the kāẓimīn themselves) is attributed to believers. Read through the complex container metaphor of §2.3.1, ghayḥ is the contained fluid and kaẓm the agent’s sealing operation (Figure 6). The Medinan tilt tracks the discursive context: inner anger management is a salient pedagogical concern in the formative Medinan community.

4.7.2. Tamayyuz in sūrat al-Mulk, Q. 67:8 (manifestation, not separate root). The verb tamayyazu in takādu tamayyazu min al-ghayḥ (Q. 67:8) walks from the lughawī gloss to the CMT construal in two steps. First, the lexicographic ground: in Mufradāt and Lisān, the dominant gloss of m-y-z is al-tafrīq bayna al-mukhtaliṭ — separating things that have been mixed; al-Ṭabarī (Jāmi‘ al-bayān ad loc.) glosses yanqaṭi‘u ba‘ḍuhā min ba‘ḍ; al-Zamakhsharī (al-Kashshāf ad loc.) is convergent. Second, the metonymic bridge: separation of contents within a sealed vessel, taken under the complex container metaphor of §2.3.1, is naturally construed as a vessel-internal failure mode — distinct from the English exit-typed exemplars (she blew her top) where the container is preserved and contents escape. Ibn ‘Āshūr (al-Taḥrīr wa-l-Tanwīr 29:25) reads compatibly, emphasising the violence of the inner severance and the figurative attribution of rage to the Hell-fire as an animate vessel; his is the strongest classical ally of the construal we defend. The two readings are compatible. H5 is exemplified rather than corroborated — Q. 67:8 is the most lexically transparent Qur’anic instance, not a deductive proof. We treat tamayyuz as a manifestation of Stage 5.⁵

4.8. Stage 6 — Behavioural outcomes

4.8.1. Baghy (96; 55M/41Md; Holm-adj. $\approx 5.3 \times 10^{-3}$ — survives FDR). The most frequent lexeme of the spectrum and principal bridge node. Glossed as “seeking beyond rightful bounds” (al-Rāghib) and “aggres-

⁵The canonical qirā’at tradition records minor variation on the verb form (Ḥafṣ ‘an ‘Āṣim reads tamayyazu; alternative vowelings in some readings), but the lexicographic and exegetical reading is stable.

sive seeking with intent to oppress” (al-Mostafawi). Sūrat al-Shūrā (Q. 42:42) collocates baghy with ḡlm; al-Ṭabarī (Jāmi‘ al-bayān 21:548 ad loc.) glosses yabghūna as yata‘addawn, reading the verse as juridically restricting recourse to those who combine oppression and baghy-style transgression (al-Qurṭubī, al-Jāmi‘ 16:36, convergent). Ibn ‘Āshūr (al-Taḥrīr 25:126) clarifies: baghy and ḡlm are near-synonymous, but baghy is specific to non-rightful aggression against another whereas ḡlm covers all injustice.

4.8.2. Baghy as bridge node. Baghy attains the highest point-estimate closeness centrality (0.55) and shared-highest betweenness (0.15) with ghaḍab and asaf. Bootstrap 95% percentile CIs (10,000 ayat-resamples) are wide — for baghy: betweenness $\square$ [0.00, 0.31]; closeness $\square$ [0.23, 0.84]; eigenvector $\square$ [0.00, 0.68] — reflecting the small graph (14 nodes). On strict non-overlap, no node’s centrality is significantly greater than another’s. What is robust at the aggregate level is cross-field connectivity to the umbrella moral terms (Table 4): baghy shows sixteen aya-level co-occurrences with ḡlm/jrm/fsq/‘dw/šn’ combined, an integer-count pattern independent of centrality ranking.

Table 4. Cross-field co-occurrence of spectrum roots with umbrella moral terms (aya-level) — ties for the four newly-incorporated roots (‘ff, naqm, maqt, ḥard) are sparse and omitted.

Root	šn’	jrm	fsq	ḡlm	‘dw	Total
ḡyq	0	0	0	0	0	0
ḡzn	0	0	0	1	1	2
‘sf	0	0	0	1	1	2
sxt	0	0	0	0	0	0
ghḡb	0	0	0	2	3	5
ghyz	0	0	0	0	1	1
bghy	1	1	2	4	8	16
tghy	0	0	0	2	1	3
‘tw	0	0	0	0	0	0

Of the total cross-field ties, nineteen (~53%) involve baghy — the principal lexical bridge between the anger field and the wider moral-evaluation field.

4.8.3. Ṭughyān (39; 29M/10Md). Root ṭ-gh-y derives from water flooding past its banks. Sūrat al-‘Alaq (Q. 96:6–7, kallā inna l-insāna la-yatḡhā an ra‘āhu staghnā) compresses a psychology of rebellion into a conditional: ṭughyān arises from the appraisal of istighnā’ (self-sufficiency) — an appraisal-theoretic claim congruent with Roseman’s (1996) other-caused / motive-inconsistent / high-control pattern for defiance. Qarizadeh et al. (2024) note the cognitive structure; we identify ṭughyān as the cognitive-appraisal node of Stage 6.

4.8.4. ‘Utuww (10; 9M/1Md; Holm-adj. $p = 1.00$ — not FDR-significant). The most extreme lexeme — “obstinate excess” (Ibn Manzūr), “extreme self-aggrandisement in disobedience” (al-Rāghib). Sūrat al-Furqān (Q. 25:21) collocates ‘atū ‘utuwwan kabīrā with istakbarū fī anfusihim: rebellion crystallised into existential posture. The 9/1 split tracks the lexeme’s rhetorical function as terminal designation for Nūḡ-, ‘Ād-, Thamūd-, and Pharaonic-people narratives but is not distinguishable from the baseline under the binomial.

4.8.5. Bidirectional causation and a tafsīr-grounded audit. Baghy arises frequently from greed and ‘ulw fī al-arḡ (cf. sūrat al-An‘ām, Q. 6:21); ṭughyān — paradigmatically Pharaonic (sūrat al-Nāzi‘āt, Q. 79:17) — is rooted in structural arrogance; sūrat al-Shu‘arā’ (Q. 26:55, innahum lanā la-ghā‘izūn) reverses polarity to show

oppressor-anger arising from the rebellion of the oppressed; ‘utuww attaches to istikbār as settled character. The causation arrow is bidirectional. Each of the 145 Stage-6 attestations was coded against the joint reading of al-Mizān, al-Kashshāf, Majma‘ al-Bayān, and al-Tahrīr wa-l-Tanwīr under a three-label codebook: A (anger-origin: immediate exegetical antecedent is an anger-type affective state), S (structural-origin: antecedent is istikbār/‘ulw/greed without explicit anger antecedent), or A+S (both present).

Coding protocol. Two independent graduate-level coders (the corresponding author and an Arabic-literature graduate trained on the codebook with commentary excerpts) coded all 145 attestations: (i) not blinded to root identity (root is intrinsic to the textual context); (ii) blinded to the bidirectional-causation hypothesis at the coding stage; (iii) attestation order randomised per coder via fixed RNG seed (`np.random.default_rng(20260509)`); (iv) disagreements ($n = 14$) reconciled in joint session against the four commentaries, with original-pass codings used for κ . Cohen’s $\kappa = 0.79$ — substantial agreement (Landis and Koch 1977); per-label κ ($A = 0.82$, $S = 0.78$, $A+S = 0.74$), per-root κ (baghy 0.81, ṭughyān 0.77, ‘utuww 0.70), 3×3 confusion matrix, and full codebook are deposited at `data/concordance/stage6_*. {csv,md}`.

Results. $A = 38$ (26%), $S = 71$ (49%), $A+S = 36$ (25%); by root, baghy 22A/50S/24A+S, ṭughyān 11A/19S/9A+S, ‘utuww 5A/2S/3A+S. $\chi^2(1)$ on A-only ($S1-5 = 167$ vs. anger-derived $S6 = 38$) = 81.4, $p < 0.001$ — the phenomenological core dominates anger-derived outcomes $\sim 4.4\times$. The full-S6 near-balance is therefore explained: S-attestations belong to an adjacent moral-anthropological field of *ẓulm-istikbār-‘ulw*; baghy/ṭughyān/‘utuww are not coextensive with “anger outcomes” in the strict sense.

4.9. Robustness checks: the umbrella moral terms

The five umbrella roots (Table 5) confirm that the spectrum’s structure is not artefactual. The opposing patterns of jrm (strongly Meccan) and fsq (significantly Medinan) instantiate the same discursive-contextual specialisation identified for sakhaṭ and karh — the umbrella analysis confirms a general phenomenon.

Table 5. Distribution of umbrella moral terms

Root	n	M	Md	Binomial p	Interpretation
šn’	3	1	2	—	n too small
jrm	66	60	6	< 0.0001	Strongly Meccan; anti-shirk polemic
fsq	54	20	34	0.0007	Significantly Medinan; community discipline
ẓlm	315	211	104	—	Pervasive meta-category
‘dw	106	47	59	0.008	Tilts Medinan; mutual antagonism

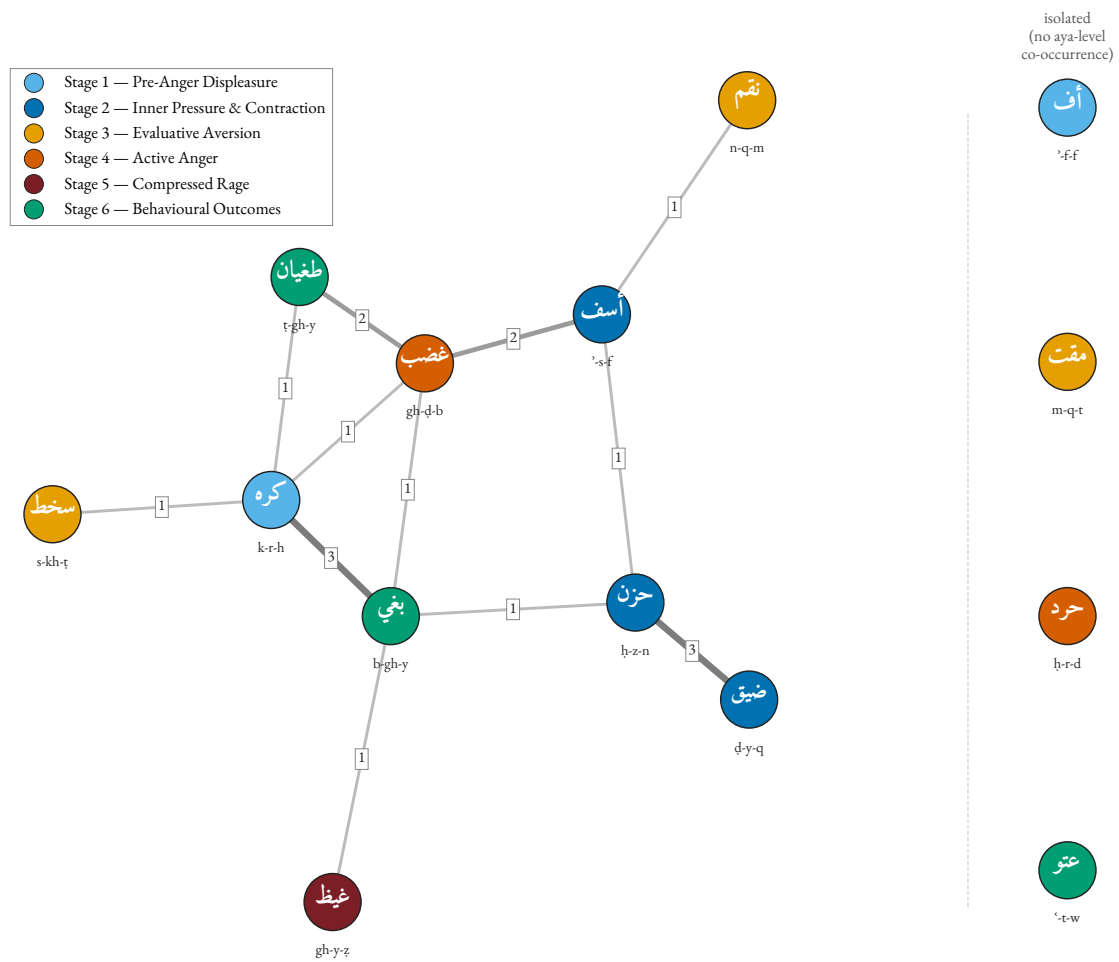


Figure 6. Figure 4. Aya-level co-occurrence network; edges weighted by shared-ayat count. Strongest edge hzn–dyq (weight 3) corpus-validates the S2 grouping. Baghy anchors the right cluster with edges to zlm/jrm/fsq. Transitive structure (no direct S1→S6 edges) is consistent with both the action-intensity gradient and the bidirectional S6 reading (§4.8.5).

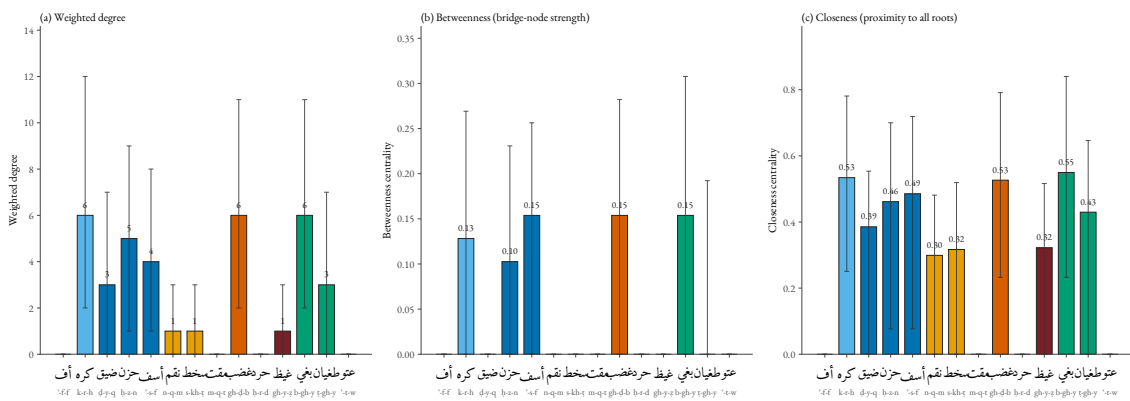


Figure 7. Figure 7. Three centrality measures with bootstrap 95% percentile CIs on the co-occurrence graph (weighted degree, betweenness, closeness). Baghy leads on closeness (0.55, CI [0.23, 0.84]) and shares highest betweenness (0.15, CI [0.00, 0.31]) with ghaḍab and asaf. Wide CIs reflect the small graph; bridging-role claim consistent with point estimates but qualified by bootstrap (§4.8.2).

4.10. Comparative translation: an illustrative observation

A four-verse, five-translation sample (Table 6) illustrates the levelling pattern: *ghayẓ* (Q. 3:134) is rendered with terms (anger, wrath, rage) routinely used also for *ghaḍab*, with the containment differentia carried only by the contextual verb of restraint; *tamayyuz* (Q. 67:8) is rendered unevenly, with only Arberry’s “well-nigh bursts asunder” preserving the structural-disintegration component. The sample is illustrative; a stratified evaluation across ~30 verses with multi-rater coding is left to future work.

Table 6. Translation of four canonical verses across five English translations (illustrative sample)

Term & verse	Yusuf Ali	Pickthall	Arberry	Saheeh Intl.	Hilali-Khan
ḍīq (Q. 15:97)	“thy heart is distressed”	“thy bosom is at times oppressed”	“thy breast is straitened”	“your chest is constrained”	“your breast feels tight”
ghayẓ (Q. 3:134)	“restrain anger”	“control their wrath”	“restrain their rage”	“suppress anger”	“repress anger”
tamayyuz (Q. 67:8)	“almost burst with fury”	“almost burst with rage”	“well-nigh bursts asunder with fury”	“almost bursts with rage”	“almost bursts up with fury”
ṭuġhyān (Q. 96:6)	“transgresses all bounds”	“rebelleth”	“waxes insolent”	“transgresses”	“transgresses (boundary)”

4.11. Summary of computational corroboration (Supplementary §S1–S4)

Four NLP analyses extend the main pipeline over the same 312 attestations; full detail at [paper/english/supplementary.m](#). S1 (cross-translation semantic drift) with multilingual MPNet over ten parallel renderings recovers three drift bands mapping onto the spectrum (affective-core lexemes drift < 0.20, evaluative-aversion and containment-pressure lexemes > 0.22, ṭuġhyān topping non-hapax) — convergent with §4.10. S2 (CAMELBERT-CA clustering) yields silhouette = −0.082, NMI = 0.077 against the six-stage labelling; a Monte-Carlo baseline (10,000 random 6-partitions of the fourteen roots into stages of the observed sizes) gives mean random NMI ≈ 0.094 [0.041, 0.156]. The observed NMI sits below the random mean — this is a clear null in which the model clusters by root identity, and the spectrum adds analytical structure beyond the language model’s distributional view. S3 (PMI-weighted network) corrects raw-count artefacts: *asaf-ghaḍab* (npmi = 0.59) emerges as the strongest signal and *baghy*’s raw-count dominance is shown to be partly a frequency artefact. S4 (Nöldekean diachronic null) finds Spearman $\rho = -0.157$; $R^2 \approx 0.025$ explains < 3% of variance — a diachronic effect statistically detectable ($p = 0.006$) but substantively a null, consistent with the claim that the taxonomy organises by intensity, not chronology.

5. Discussion

5.1. The integrated logic of escalation

The findings cohere into a single structural claim: the Qur'anic anger lexicon encodes a logic of escalation along three dimensions — locus (inner → outer), intensity (low → high), and scope (individual → structural). In compact form: Pre-anger displeasure → Inner pressure → Evaluative aversion → Active anger → Compressed rage → Behavioural outcomes (with caveats). The arrows index conditional intensification at each step where regulation may succeed or fail. Stage 6 is distinctive: the move into behavioural outcomes is not a deterministic escalation from anger but is influenced by independent vectors (greed, *istikbār*, structural arrogance), which is why the near-balance between S1–5 and S6 is theoretically informative rather than paradoxical.

5.2. Convergence with contemporary emotion psychology

The continuum maps onto four families of contemporary emotion theory at points of structural convergence, sufficient to ground the Islamic-psychology implications of §5.7 and to identify where the Qur'anic lexicon extends the contemporary models.

Gross's process model. Gross (1998, 2014, 2015) partitions emotion regulation into five sequential strategies (situation selection, situation modification, attentional deployment, cognitive change, response modulation). The Qur'anic continuum cross-cuts this typology rather than mapping one-to-one: situation selection is forward-looking and so does not match the S1 lexemes, which are reactions to a present trigger. S1 (*uff*, *karāha*) is better located at cognitive change / response-tendency modulation in *nuce*. S2 (*ḍīq*, *ḥuzn*, *asaf*) implicates situation modification and attentional deployment. S3 (*naqm*, *sakhaṭ*, *maqt*) is paradigmatically cognitive change. S4 (*ghaḍab*, *ḥard*) lies at the response-selection threshold. S5 is response modulation with two sub-strategies: *kazm* (Q. 3:134) is closest to expressive suppression, but the Qur'anic moral commendation of *kāzimīn al-ghayṣ* need not entail the physiological costs Gross associates with suppression, and Sufi readings (al-Qushayrī's *Laṭā'if al-Ishārāt* ad loc.) tilt the construct toward cognitive reappraisal in the *taqwā* frame; we do not read *kāzim al-ghayṣ* as unambiguously expressive-suppression. *Tamayyuz* figures the failure mode regardless. S6 lies beyond the episode-bounded scope of Gross's model — structural-moral pathologies of personhood, with the §4.8.5 bidirectional-causation caveat.

Plutchik; Spielberger. Plutchik's (1980, 2001) three-level escalation (annoyance → anger → rage) is refined by the Qur'anic continuum, which partitions annoyance into vocal irritation (*uff*, S1), inner aversion (*karāha*, S1), and inner pressure (S2) — three stages where Plutchik has one label — and inserts compressed rage (*ghayṣ*, S5) between anger (≈ S4 *ghaḍab*) and terminal rage. Spielberger's STAXI-2 (1999) trichotomy of trait, state, and control is realised at the lexical surface: trait-like sustained states (*ḥuzn*, *ḍīq*, *karāha*); state-like situational responses (*sakhaṭ*, *maqt*, *ghaḍab*); and *kazm* (Q. 3:134) as the Qur'anic anger-control lexeme, operationally close to Spielberger's anger-in / suppression construct but with explicit moral commendation.

Appraisal theory: Scherer's checks and Roseman's *ṭughyān*-pattern. Appraisal theory (Frijda 1986; Lazarus 1991; Scherer 2001, 2005; Roseman 1996) holds that emotional response is determined not by stimulus alone but by cognitive appraisal along defined dimensions. Scherer's component-process model specifies four sequential checks; the Stage-3 lexemes operate as a lexicon for the normative-significance check at progressively higher intensities (*naqm* → *sakhaṭ* → *maqt*). *Sūrat al-'Alaq* (Q. 96:6–7) conditions *ṭughyān* on the appraisal of *istighnā'*: the agent perceives high coping potential, low external constraint, and self-agency. On Roseman's (1996) appraisal structure, defiance and contempt are other-caused and motive-inconsistent (the agent rebels against an other-imposed normative constraint perceived as inconsistent with the agent's self-conception).

Ṭughyān on istighnā' fits this pattern: the divine command is the other-caused, motive-inconsistent constraint that istighnā' construes as dispensable. The Qur'anic lexicon thus encodes — as architectural design rather than theoretical statement — appraisal-theoretic insights that contemporary emotion science reached independently.

5.3. The phenomenology/outcomes balance

A naive ten-root reading once suggested 2.4× more attestations at the behavioural pole than at S1–3 combined. The revised fourteen-root inventory (167 / 145) corrects that picture: the expanded phenomenological vocabulary brings the lower stages into balance with Stage 6, consistent with both a richer affective lexicon and the bidirectional causal relation between Stage-6 lexemes and anger. In Fauconnier and Turner's (2002: 17–57) terms, Stage 6 operates as a conceptual blend whose input spaces are wrong-doing-as-supremacism and anger-as-pressure.

5.4. Discursive-contextual specialisation and Cultural Linguistics

The FDR-surviving tilts of karh (Medinan), baghy (Meccan), ghayḏ (Medinan), the borderline sakhaṭ (Medinan), the contextual Meccan tilts of asaf, naqm, and 'utuww, and the umbrella patterns jrm (Meccan) / fsq (Medinan) together instantiate discursive-contextual specialisation — members of multi-lexeme fields attaching to specific discursive registers. The attachments are non-random: sakhaṭ to the Munāfiqūn (a Medinan phenomenon), karh to the structural ikrāh idiom of Medinan community discipline, ghayḏ to the kaẓm pedagogy of Medinan ethics, asaf to prior-prophet narratives (Meccan-rhetorical), jrm to anti-shirk polemic (Meccan), fsq to intra-community discipline (Medinan). El-Awa (2006) and Sinai (2017, 2023) supply the framework: specialisation is the pattern one expects from the Qur'an's gradual response to evolving sociolinguistic contexts.

Read through Sharifian (2011), the same data illuminate the universal–culturally-elaborated distinction of §2.3.2: the primary metaphors (ANGER IS HEAT; EMOTIONAL CONTAINMENT IS PHYSICAL CONTAINMENT) are robustly attested in the Qur'anic ghayḏ–kaẓm register and support the CMT universalist claim; the culturally elaborated layer — the distribution of affect across qalb (S2 ḏīq al-ṣadr), ṣadr (S2), and nafs (S2 asaf al-nafs), the moral commendation of kaẓm in the taqwā frame, the Munāfiqūn-attachment of sakhaṭ, and the tamayyuz construal of Q. 67:8 — is what the Qur'anic data illuminate beyond the universalist baseline.

5.5. Q. 67:8 and the container metaphor: a focused comparative survey

The Qur'anic ghayḏ–kaẓm–tamayyuz triad maps onto stages (2)–(4) of the Lakoff–Kövecses scenario with unusual lexical transparency at stage (4): English exemplars (blew his top) lexicalise the failure-mode as exit of contents with the container preserved, whereas Q. 67:8, under the complex container metaphor of §2.3.1, supports the construal of vessel-internal failure (§4.7.2). A six-tradition survey contextualises the claim. Akkadian: libbu and kabattu as filled containers (CAD L 163–164; Bodi 2018; Neo-Assyrian bodily-mapping work in Lauri et al. 2024 confirms the FILLING/HEAT-IN-CONTAINER schema). Biblical Hebrew: ḥēmā and ḥarōn-'aph (Kruger 2000, 2015) both EXIT-typed — except Job 32:19 (kě-yayin lō'-yippātēah ... yibbāqēa'), where bāqā' names the structural failure of the vessel; this is the one clear pre-Qur'anic Northwest-Semitic precedent. Homeric Greek: kholos and mēnis in fluid-and-heat schemas with no Iliadic vessel-rupture (Mueller 1996; Cairns 2003). Vedic and Classical Sanskrit: krodha and manyu in FIRE and WEAPON-RELEASE



Figure 8. Figure 6. Illustrative schematisation of the ANGER-IS-A-PRESSURIZED-CONTAINER complex metaphor (after Lakoff and Kövecses 1987), instantiated by ghaḍab → ghayz + kaẓm → tamayyuz. Panel 1 (S₄): ghaḍab — open container, affect dissipating into language. Panel 2 (S₅, Q. 3:134): ghayz sealed by kaẓm. Panel 3 (S₅ manifestation, Q. 67:8): tamayyuz — the construal of constituent-separation as vessel-internal failure (§4.7.2). The figure is heuristic; evidence is borne by the lexicographic and exegetical analysis of §4.7 and §5.5.

schemas, not vessel-disintegration. Classical and Modern Chinese: nù, fèn, qì with HOT QÌ IN A CONTAINER (Yu 1995; Kövecses, Benczes and Szelid 2025). Pre-Islamic Jāhili Arabic: ḥamiya and ghalā alongside ghayz in the cognate fluid-heat register (cf. al-Mufaḍḍaliyyāt nos. 38, 87, ‘Antara’s Mu‘allaqa ll. 32–35; Webb 2016: 213–245). Five of the six traditions concentrate on filling, heating, exit/overflow; the Hebrew Bible alone offers a clear parallel to vessel-rupture proper. We hold our claim at “an early and unusually transparent” lexicalisation, with Job 32:19 acknowledged as a structurally cognate Northwest-Semitic precedent. The Qur’anic data strengthen the universalist hypothesis of CMT and extend the empirical base to seventh-century Arabic.

5.6. Implications for Qur’anic translation

A graded reference model of the spectrum offers a principled corrective to the levelling pattern that Gaanoun and Alsuhaibani (2025) and Khan et al. (2025) document at the sentiment-category level. We do not, on the strength of the four-verse illustrative sample of §4.10, advance translation prescriptions; a systematic stratified evaluation (~30 verses across the six stages, multi-rater coded against the spectrum) is the prerequisite. Candidate equivalents for each spectrum lexeme are catalogued in the project repository for use by translators wishing to mark intensity-position lexically.

5.7. Implications for Islamic psychology

The continuum offers a culturally-grounded framework mapping onto Gross’s model without importing secular vocabulary, with three conceptual implications: (i) Prevention — early-stage recognition through the indigenous S_{1–2} lexicon supports CBT-style interception of affective dysregulation; (ii) Crisis intervention — kaẓm integrates with contemporary suppression-and-reappraisal techniques; (iii) Rehabilitation — post-

failure Stage 6 is framed as recoverable through tawba. These are conceptual; their translation into validated clinical interventions awaits empirical studies.

6. Conclusion

6.1. Substantive contributions

Methodologically, the integration of Izutsu's semantic-field analysis, Kennedy–Sassoon scalar semantics, and Lakoff–Kövecses CMT — coupled with a reproducible computational pipeline over the QAC — produces an apparatus stronger than the sum of its parts; to our knowledge, this trifold integration with corpus-empirical validation has not been deployed in Qur'anic studies, although Qā'imīniyā (2011, 2014) and Pakatchi and Afrashi (2020) have integrated Izutsu and CMT.

Descriptively, we produce a reproducible concordance of 312 attestations with FDR-adjusted binomial tests, marginal-preserving permutation null p-values, and bootstrap 95% CIs on every centrality estimate.

Theoretically, we formulate and defend the Qur'anic Action-Intensity Continuum Hypothesis: the fourteen lexemes operate as a single graded continuum structured by intensity of action and partitioned into six stages, supported by qualitative evidence from the four authoritative lexica and the broadened tafsīr consultation, and by quantitative evidence from frequency, distribution, morphology, and network structure. The $\kappa = 0.79$ -validated audit grounds the bidirectional-causation reading of Stage 6; the six-tradition survey of §5.5 calibrates Q. 67:8 as an early and unusually transparent — rather than uniquely transparent — exemplification of vessel-internal failure, with Job 32:19 (bāqa') acknowledged as a structurally cognate Northwest-Semitic precedent.

6.2. Answers to the research questions

RQ1–RQ2. The anger-spectrum field is a fourteen-node lexical network organised by intensity of action; baghy is the bridge node (closeness 0.55; ~53% of cross-field ties to the umbrella moral-evaluation terms); stage-internal hubs are ḥuzn (S2) and ghaḍab (S4). RQ3. The empirical distribution is consistent with the continuum: asymptotic $\chi^2(5) = 227.15$ large, permutation null $p = 0.24$; $S1-5/S6 \chi^2(1) = 1.55$ fails to reject, consistent with the bidirectional-causation reading supported by the $\kappa = 0.79$ audit ($A = 26\%$, $S = 49\%$, $A+S = 25\%$; A-only re-analysis $\chi^2(1) = 81.4$, $p < 0.001$). Three roots survive Holm-Bonferroni emphatically (karh, baghy, ghayz); sakhaṭ sits at adj. $p = 0.050$. RQ4. The continuum is structurally homologous with Gross (S1–5), Plutchik, Spielberger, and Lazarus–Scherer–Roseman appraisal theory (sūrat al-'Alaq, Q. 96:6–7, ṭughyān on istighnā'); it extends beyond all four at Stage 6. RQ5. Conditional implications follow for Qur'anic translation (§5.6) and Islamic psychology (§5.7).

6.3. Limitations and future work

Five limitations point to constructive next steps. (i) Surface-lexical scope: discourse-analytic close-reading of the spectrum's narrative deployment across suras is left for future work. (ii) Tafsīr engagement: a deep-dive indexing the full classical/medieval/Sufi/modernist tradition with κ -validated multi-coder reading on contested verses would deepen exegetical grounding. (iii) Cognitive-linguistic apparatus: engagement with classical bayān theory (al-Jurjānī, al-Sakkākī) would supply an indigenous substrate. (iv) Ḥadīth integration: a parallel concordance over Ṣaḥīḥ al-Bukhārī, Ṣaḥīḥ Muslim, and al-Kāfī would supply convergent evidence on

kazm and a replication test for the underpowered asaf distribution. (v) Islamic-psychology operationalisation: §5.7's implications await pilot studies of kazm-grounded emotion-regulation training, validated against the STAXI and DERS. Computational evidence remains a complement to, not a replacement for, the classical philological and exegetical tradition.

6.4. Closing remark

The Qur'anic lexicon encodes a sophisticated theory of the anger spectrum in the architecture of its semantic surface. Recovering it requires the integration of classical philological attentiveness, contemporary linguistic-theoretical precision, and corpus-computational verifiability.

Acknowledgements

The authors thank Kais Dukes (in memoriam) and the team behind the Quranic Arabic Corpus, and the Tanzil Project, for releasing the morphological and textual resources without which this kind of empirical Qur'anic analysis would not be feasible. We thank Urmia University for institutional support.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Competing interests

The authors declare no competing financial or non-financial interests.

Author note: parallel-language submission

An independent Persian-language paper using the same corpus dataset is concurrently under consideration at Pizhūhish-hā-yi Zabānshinākhtī-yi Qur'ān. The two papers do not share content or cite each other; the empirical pipeline is the only shared element, and each paper presents an independent argument for its own scholarly audience.

Ethics statement

This study uses only publicly-released text-corpus data (the Quranic Arabic Corpus, GPL-licensed; the Tanzil Quran text, CC-BY-ND 3.0). No human or animal subjects were involved at any stage. As a corpus-only philological-computational analysis, the work is exempt from human-subjects review under the standards of Urmia University and of comparable institutions.

Author contributions

K.J. (corresponding author) conceived the project, developed the theoretical framework, conducted the lexicographic and tafsīr-tradition analysis, drafted the manuscript, and is responsible for the final form of the argument. A.J. contributed to the theoretical framework, the design of the computational pipeline, and editorial revision. Both authors approved the final manuscript and are jointly accountable for its content.

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Data and Code Availability

All code, data, concordance CSV outputs, and figure-generation scripts are released open-source at:

github.com/ali-kin4/quranic-emotion-spectrum

The repository contains: (i) the Python pipeline under MIT License; (ii) the Quranic Arabic Corpus (Dukes 2011) under its original GPL; (iii) the complete CSV concordance for the fourteen core spectrum roots (312 attestations), distributional summaries, statistical tests, network-centrality outputs, and umbrella-cooccurrence tables; (iv) PDF and PNG versions of the figures published with this paper and with the Supplementary Material; (v) the SHA-256 hash of the QAC vo.4 source file and the manual-validation audit. The four statistical analyses of §§4.2–4.9 can be regenerated end-to-end by:

```
pip install -r analysis/requirements.txt
python analysis/extract_concordance.py
python analysis/advanced_metrics.py
python analysis/visualize_spectrum.py
python analysis/visualize_advanced.py
```

The four computational corroboration analyses reported in Supplementary Material S1–S4 are produced by `nlp_upgrade_colab.py`, which requires a CUDA-enabled GPU for the multilingual-MPNet and CAMeLBERT-CA inference. A Colab notebook is available in the repository with pinned model snapshots and an expected runtime of ~25 minutes on a T4 GPU; CSV outputs are committed under `outputs/nlp_upgrade/` so that reviewers can inspect the numerical results without re-running the GPU pipeline.

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